EPSR-test-ex3 Unbiased automatic ereq stepping (bias 0)

Program(s): epsr | chained passes: 6 | verification: ALL IDENTICAL (all 6 passes)

What this case tests

Exercises the unbiased branch of EPSR's automatic EP-amplitude control [1,2,3]: with `bias 0` the step of `ereq` is drawn at random in both directions rather than being pushed along the quality-factor trend, which tests the random-step formula and the rounding of `ereq` to multiples of the step unit.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` lines 2180-2262 (automatic `ereq` step, zero-bias branch), reading of `ereqstep/bias`
- `epsr_standalone_rev3.f`: `lsparabola1` (trend line of the `.erg` history), `ptgnsq`
- `auxiliary_routines.f`: `ran1`; `read_and_save_files.f`: `save_files_others`

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::automatic_ereq`, `Epsr::setup`
- `potential.rs`: `lsparabola1`, `State::ptgnsq`; `rng.rs`: `Ran1`
- `pcof.rs`: `save_pcof_file`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. The EP amplitude starts at `ereq` = 10 kJ/mol and is stepped by 2 kJ/mol without bias.

Inputs

Copy of EPSR-test with only the `.pcof` header changed: `ereqstep 2.000 1.000` (0 in EPSR-test) and `bias 0` (2 in EPSR-test).

How the accuracy was verified

`commands.txt` (1 program call, repeated 6 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 6 passes.
Six chained passes with feedback of `.pcof` and `.erg`. The unbiased stepping is visible in the Fortran reference: the EP energy column of the six new `.erg` lines reads 10, 10, 12, 12, 10, 10 and the final `.pcof` has `ereq` = 8, i.e. steps both up and down. Since each step consumes `ran1` random numbers, identical output also confirms that the port draws them in the same order. All 27 output files are identical after every pass.

Notes

Fortran behaviour reproduced: the second number of `ereqstep` is ignored on input (the unit is set equal to the step), so the file is written back as `ereqstep 2.000 2.000`.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.